

Proposed Solution

Date	3 July 2026
Team ID	SWTID-2026-8260
Project Title	A Comprehensive Measure of Well-Being using HDI Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	To develop a machine learning-based system that predicts human development index (HDI) and provides insights into overall well-being using multiple socio-economic factors.
Scope	• Data preprocessing and analysis • Model training and evaluation • Web application development • Visualization of insights • User input-based predictions
Problem Statement	
Description	Existing systems fail to provide a comprehensive and predictive analysis of well-being. They rely on static reports instead of real-time intelligent systems.
Impact	• Poor decision-making • Lack of predictive insights • Inefficient policy planning
Proposed Solution	
Approach	• Use machine learning algorithms for prediction • Train model on HDI-related datasets • Build an interactive web interface using Flask
Key Features	• HDI prediction based on inputs • Data visualization (graphs, heatmaps) • User-friendly interface • Real-time results

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Used for running ML models and processing data	CPU / Basic GPU
Memory	Required for handling datasets and model execution	8 GB RAM
Storage	Stores datasets, trained models, and outputs	1 TB SSD
Software		
Frameworks	Used to build backend web application	Flask
Libraries	Used for data processing, visualization, and ML	pandas, numpy, sklearn, matplotlib, seaborn
Development Environment	Tools used for coding and testing	Jupyter Notebook, VS Code
Data		
Dataset	Provides input data for training and testing model	HDI dataset (CSV format from Kaggle)